Hotel Booking Cancellation Analysis

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***Abstract*—Hotel cancellations create huge losses of money to hotels, to prevent this we must understand why they happen in the first place.**

This study aims to identify the key drivers of hotel booking cancellations. It focuses mostly on seasonality, customer differentiation, hotel fee and lead time analysis. These questions are crucial in hospitality management and churn analysis. Descriptive analysis, T test, ANOVA and linear regression methods are applied to better understand the dataset. It is found that monthly booking volume, familial status and lead time do influence hotel booking cancellations.

***Index Terms*—Data Analysis, Data Visualization, Hotel Booking, Hospitality Management**

I. Introduction

A. Overview

What causes a person to withdraw from a dreamed vacation at a hotel? Each person has their own answer to this question and personal reasons come into play. However, it might be possible to pinpoint some common elements about customers who cancel their booking. This paper aims to find factors that influence cancelation of a hotel booking by analyzing real data from two hotels in Portugal.

For this, this paper will first focus on a descriptive analysis on variables, this will include visualization to better understand the trends about customers. From our findings, we will infer results about general population. We will use common data analysis methods to better understand our data and to perform an accurate analysis. After comparing our results, we will continue with the method which will suit this data the best.

B. Motivation

This study is mainly conducted for personal interest, the author is by no means a hotel employee. Yet, it is written from the viewpoint of one, as the end result can be interpreted for marketing and customer segmentation projects. This segmentation can be done like ”groups of people who cancel the most/least their booking”. With these findings, hotels can further their marketing for these customer segments in different ways. For example, if we find out at the end of this research that people who book who have babies cancel their booking the most, hotels can promote more baby-friendly environments for this customer type.

C. Research Questions

With this dataset, there is much to discover and many have used it to make prediction models. [3] This study will focus specifically on the attributes that affect hotel booking cancellations. Our main question will be ”What are the factors which influence the cancellations the most in these specific two hotels in Portugal, between the years 2015-2017?” Of course we can only give an answer within the limits of our dataset.

Other research questions can be listed as : ”Cancellations happened mostly for which time periods (month/week number)?”, ”Do people with babies and/or children cancel more often their booking?”, ”Is there a relationship between the fee of a hotel and cancellations?”, ”Which kinds of hotel rooms tend to be cancelled more?”, ”Can we observe a relation between the cancellation rate and how the booking was made?”

In short, this study can be explained by ”churn analysis in hospitality”. In this context, customers who cancel are the ones who churn. And we ask why our customers churn.

D. Contributions

This research provides a quantitative framework to prove that hotel cancellations are not random, but based on connected data.

A primary technical contribution of this work is the feature engineering of the ”Family Status” variable. Through statistical T-tests, we identified a higher ADR for this segment, while ANOVA results confirmed their significantly lower cancellation rates. This mathematical evidence challenges the idea that higher prices always lead to higher churn, establishing that families represent a ”low-risk, high-revenue” customer type. Hotels can promote towards this category for more revenue.

This study found a Pearson correlation coefficient of $r = 0.742$ between monthly booking volume and cancellation rates. This correlation proves that as the hotel scales, the probability of cancellation increases.

Also, we found that each special request acts as a negative weight on the cancellation probability. This proves that guests want to invest in an experience that creates a psychological commitment.

II. Method

A. Dataset

1) Story/Overview: This dataset consists of real data, took from two hotels in Portugal. One is a resort at Algarve and the other is a hotel at the city of Lisbon. Data is collected over the span of two years, between summer of 2015 and 2017. It should be noted that these dates are before Covid-19 which effected the tourism sector greatly. Also, online booking became more common after this data has been published [I

need to find a ref for this].

Dataset I encountered at Kaggle website is actually taken from an article [1]. In the article, the team explains that this dataset has an important role for data analysis purposes, given that real data from real businesses are scarce in hospitality field. They collected this data based on 40,060 observations from the resort and 79,330 from the hotel. Therefore, there are over 100,000 separate columns in this dataset, this creates a reliable source for our analysis. One problem that might occur is that our data is limited to only two sources: two hotels. So what we encounter here might not be accurate for each hotel in the world, but might only be the case for some hotels. Further comparisons with other data should be made for testing accuracy.

2) Attributes: There are 36 features in this dataset downloaded from Kaggle. The original one, which came from the paper, had 31. Between the added five, four of them are artificially created customer information. Our research will not be considering these features important and they will be ignored. The other is "hotel" data, which indicates from which hotel we acquired the features.

Other attributes, their data type and detailed explanation are in this table below :

Our target variable is "is canceled".

Arrival date is excluded, all data is between years 2015-2017. We want to look for seasonal trends that stand true no matter the year.

"Market segment" means what type of customer made the booking. It can be a TA (Travel Agency user) or a TO (Tour Operator user). There are also rarer values such as direct, corporate and groups. It also gives information about whether booking is made online or offline. "Distribution channel", on the other hand, means if booking is made by a TA (Travel Agency) or a TO (Tour Operator) or by some other method.

The distribution channel can be connected with "agent" attribute. It gives us the ID of whichever agency did the booking. If there were no agents, there is a null value instead. Before using the dataset, these null values should be cleared.

"Company" is not to be confused with "agent", as it indicates clients who are companies themselves. There are also many null values to be cleaned here.

ADR is one of the complex variables here as it is acquired by a formula :

$$ADR = Totalroomrevenue/Totalnumberofstayingnights$$

This variable gives us an idea about expensive the booking is.

TABLE I
ATTRIBUTE DESCRIPTION

Attribute	Type	Description
hotel	categorical	Resort or city hotel
is canceled	binary	If booking was canceled (1) or not (0)
lead time	interval	Number of days between booking and check-in date
arrival date month	categorical	Month of arrival
arrival date week number	interval	Week number of arrival
arrival date day of month	interval	Date of the day of entry
stays in weekend nights	ratio	Number of overnight stays per weekend days booked
stays in week nights	ratio	Number of overnight stays per week days booked
adults	ratio	Amount of adults
children	ratio	Amount of children
babies	ratio	Amount of babies
meal	categorical	Type of meal booked
country	categorical	Which country customer is from
market segment	categorical	Market segment designation
distribution channel	categorical	Booking distribution channel
is repeated guest	binary	Customers who have stayed at this hotel before
previous cancellations	ratio	Number of previous canceled bookings
previous bookings not canceled	ratio	Number of previous non-canceled bookings
reserved room type	categorical	Booked room type (what costumer wants)
assigned room type	categorical	Given room type (what hotel assigned)
booking changes	ratio	Number of changes made to the booking
deposit type	categorical	Is there a deposit to guarantee the booking
agent	categorical	ID of travel agency
company	categorical	ID of company that made the booking
days in waiting list	ratio	Number of days passed before booking confirmation
customer type	categorical	Type of customer
adr	interval	Average daily rate
required car parking space	ratio	Number of car parking spaces
total of special requests	ratio	Number of special requests
reservation status	categorical	Current reservation status
reservation status date	categorical	Date where last status was set

B. Method for Answering Research Questions

Before starting our analysis, it is checked to see whether there is any need for data preprocessing. It is found that there were no duplicate data. Some attributes had null values (4 in children, 4488 in country, 16340 in agent and 112593 in company). We drop country nulls as they do not make much of our data, and change others with 0. By creating a box plot for every attribute with a non-categorical data, we can see outliers. We exclude them from our dataset. For some visualizations, categoric data (months) have been mapped into numeric data.

For temporal analysis, we will be using ANOVA to separate months and compare them between each other. After

ANOVA, a post hoc analysis will be used to better see the specific differences between groups.

For investigating correlations to cancellation rates, we will use Pearson correlation coefficient and then try a simple linear regression.

Independent variables T-test will be used in various places to understand the significance of difference between variables.

Analyses were performed using Python with these libraries : Pandas for data manipulation, SciPy for statistical testing, and Seaborn/Matplotlib for data visualization

III. Results

A. Descriptive Analysis

[34]:

	is_canceled	lead_time	arrival_date_year	arrival_date_week_number	arrival_date_day_of_month	stays_in_weekend_nights	stays_in_week_nights
count	118902.000000	118902.000000	118902.000000	118902.000000	118902.000000	118902.000000	118902.000000
mean	0.371373	104.308027	2016.157617	27.166726	15.800567	0.928891	2.502136
std	0.483174	106.903127	0.707479	13.589774	8.780371	0.996211	1.900165
min	0.000000	0.000000	2015.000000	1.000000	1.000000	0.000000	0.000000
25%	0.000000	18.000000	2016.000000	16.000000	8.000000	0.000000	1.000000
50%	0.000000	69.000000	2016.000000	28.000000	16.000000	1.000000	2.000000
75%	1.000000	161.000000	2017.000000	38.000000	23.000000	2.000000	3.000000
max	1.000000	737.000000	2017.000000	53.000000	31.000000	16.000000	41.000000

Fig. 1. By using 'describe' function, we can see interquartile ranges, mean, standart deviation etc.



Fig. 2. Our target variable : cancelled bookings. The amount of canceled bookings is more than 40,000



Fig. 3. Examples of histograms. There is a surge of bookings made for 27th week. Many bookings were for 0-2 weekend nights and 0-8 week nights.

```
q_high = df['adr'].quantile(0.9999)
df_clean = df[df['adr'] < q_high]

sns.boxplot(x='is_canceled', y='adr', data=df_clean)
plt.title('ADR: Canceled and Not Canceled')
plt.show()
```

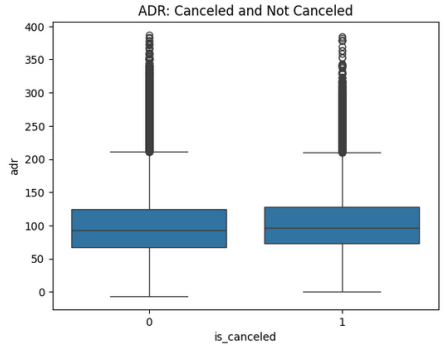


Fig. 4. There was an outlier at ADR attribute. By removing the top 0.01% percent, we can see the outlierless boxplots. There seems to be a slight correlation between high ADR (near mean) and cancellation rate.

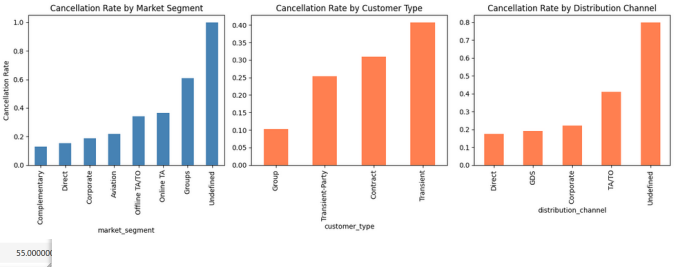


Fig. 5. We can divide our categorical data with the help of bar charts. These charts give us insight about which kinds of customers/booking types mostly resulted in cancelled bookings.

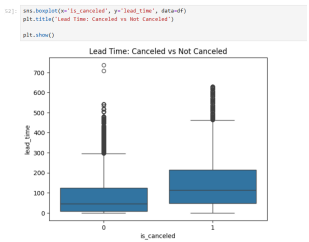


Fig. 6. After this box plot, we can deduce that there is a correlation between longer lead time (number of days between booking and check-in) and canceled bookings

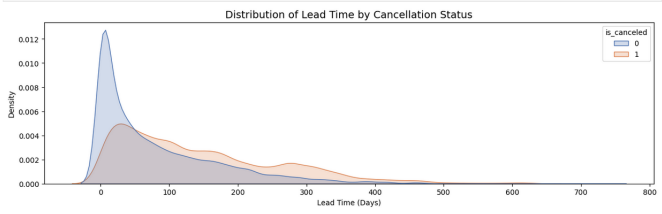


Fig. 7. Here we can see clearly that the longer lead time is, more cancellation happens.

One attribute had caused confusion : ”reservation status date” does not mean the date for which the booking was made for. It gives us insight about when the last status is set. To better understand how it works, we can give an example. A person who booked a hotel for december might have canceled the booking at august, but also there were people who checked out at august; therefore all data related to these months get combined. Attempts to interpret results about time related data with it would result in misinterpretation.

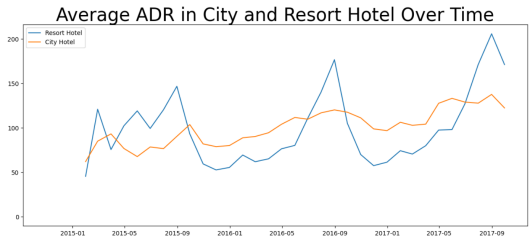


Fig. 8. Here are average daily rate (ADR) data, shown in a line graph over time of last status was saved. The time series here cannot be used to find correlations about hotels and their ADR. Because we do not know if ADR corresponds to a canceled booking (reserved for a future date) or an already checked in room.

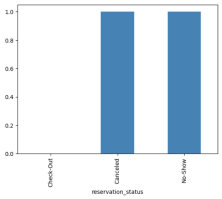


Fig. 9. Reservation status data has 3 categories. No-Show : did not actively do a booking cancelation, they simply did not show up. Canceled : canceled their booking prior to arrival date

Attribute we are searching, ”is canceled” do not take into account that these two are different and puts them together.

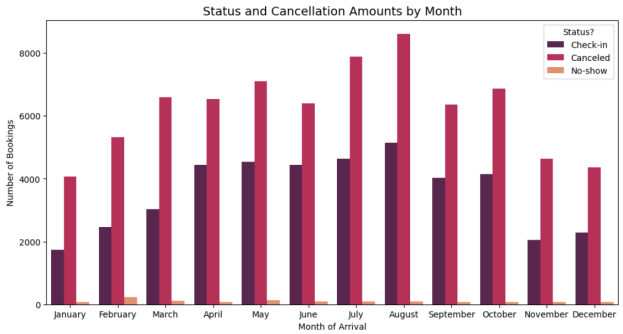


Fig. 10. This plot shows us how much of each status occurred on a monthly basis. No-show amount is very small compared to other status.

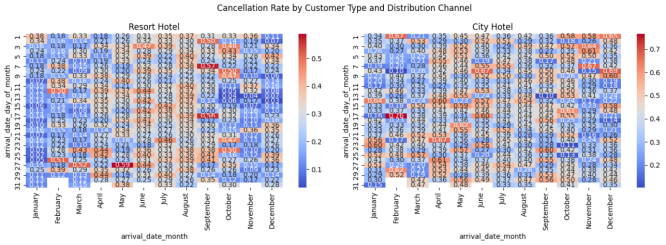


Fig. 11. These heatmaps show us every single day of a year and the rate of cancellation. We can observe a high rate of cancellation for summer bookings for resort (April-September). City hotel bookings are more dispersed but April-May bookings have a higher rate of cancellation.

B. *Analysis for Answering Research Questions*

After doing a detailed descriptive analysis, we have enough evidence to move on to specific questions. First question we are handling is : ”Cancelations happened mostly for which time period?”

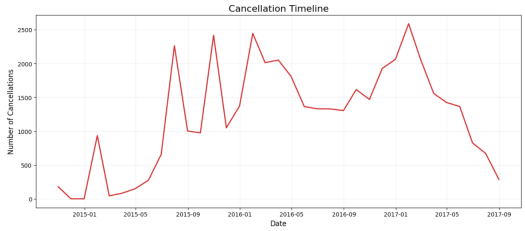


Fig. 12. This chart answers the question ”When do people cancel their bookings? (No-Show is not included, only cancelations prior to booking)”

This chart does not represent our research question. We will not be focusing on ”when” the cancelation happened, but ”for when” it was made and later canceled (intended arrival time).

Based on the heatmap of each day of the month (Fig. 11): We can hypothesise that there is a difference between cancelation rates of months. Our H0 is, ”All months have the same cancelation rate.” H1 is, ”At least one month have a different cancelation rate than others.”

$$Ho : \mu_1 = \mu_2 = ... = \mu_{12}$$

$$H1 : \exists i, j : \mu_i \neq \mu_j$$

We conduct one way ANOVA by grouping each month and looking at the same days for each month. Our F critical value is around 1.82. ANOVA proves us that our H0 hypothesis is false, as F value seems to be 6.39. And accordingly, p value is very minimal : around 1.1e-09. This is far smaller than 0.05, the alpha value we considered, so it is very significant. This exceptionally low p-value is a result of both the strong patterns in the dataset and the large sample size.

To identify where groups have differences, we have to conduct a post hoc analysis. By using Tukey’s Test, we can see between which months there is a significant difference between the means of cancellation rates.

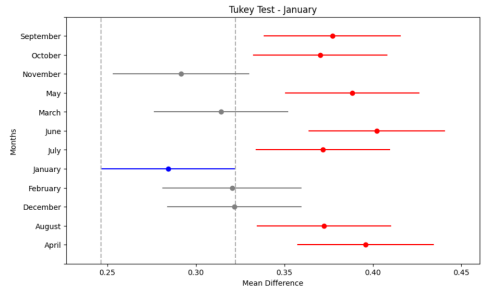


Fig. 13. Difference between January and other months. Between April-October (shown in red), there is a significant difference. These months have much higher rate of cancellation than January has.

Is the fact that summer months have higher rates of cancellation merely a correlation with summer months having overall more reservations? As we are working with rates, the answer is no : rates are independent of population size. So there seems to be a difference in the behavior of different season hotel guests.

However, we can go on and ask another question : "Is the summer months having more people canceling rate related to the fact that reservation volume is high?" For answering this question, we need to see if there is a correlation between total booking amount and cancellation rate. Pearson correlation coefficient for it is 0.742. This is a little lower than 0.8 so there is a slight correlation.

With the help of a scatter plot, we observe that there is a linearity. It is not very homogeneous, but it still gives us an idea about reservation amounts effect on cancellation rate.

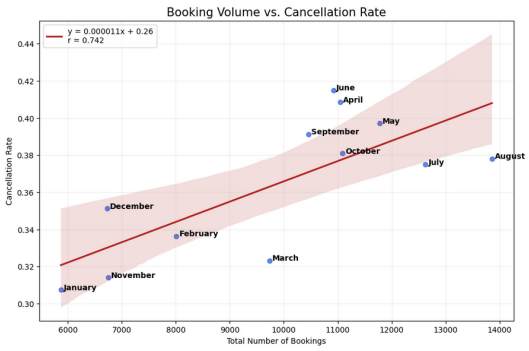


Fig. 14. Hotel can use this linear equation. Slope signifies how risk of canceled rooms is scaled by volume of bookings.

For analyzing whether families with children/babies do cancel more or less, we should take a similar approach. Here we will be using a T test (because we do not know the population variance, hence we cannot use Z test).

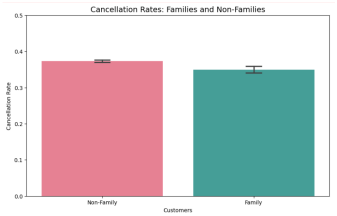


Fig. 15. There seems to be a slight difference between families and childless people when it comes to canceling hotel bookings.

In our cleaned dataset, where we eliminated some outliers from the adults/children/babies attributes, the size of the family data is 9315 (the number of instances where adults came to the hotel with children; this does not necessarily mean family, but we will call it so for simplicity). Adults who booked without children make up 109570 instances. From Fig 15, we hypothesize that family does affect cancellation rates:

$$Ho : \mu_f = \mu_n$$

$$H1 : \mu_f \neq \mu_n$$

Our degree of freedom is more than 100.000. We calculate T test result, it comes out as -4.48. The minus here shows that our first variable (family) have a lower cancellation rate than the second (non family). Based on Fig 15, this is correct.

p value is less than 0.00001. Therefore probability that H0 stands true is very low (lower than 0.05), in this case we conclude that H0 is rejected. Family status is a predictor of cancelation behavior.

What else can we deduce from family and non-family groups?

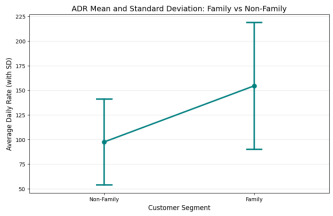


Fig. 16. Family customers seems to do higher priced bookings.

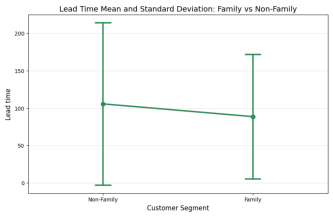


Fig. 17. Non family customer type do have longer lead times on average, and have a greater standart deviation.

Also proved by T tests (where p were extremely small), we can say that H0, "ADR/Lead time do not depend on familial status", is no longer recognized.

This means, family instances mostly selected high value hotels with lesser waiting time compared to non-family counterparts. This information is important for hotels : Family customers are more valuable and they cancel less.

Fig 4 showed us that canceled bookings have a slightly higher ADR rate on average. We calculate that mean ADR for canceled bookings are 104.94 while mean ADR for non canceled bookings is 100.21. Despite the difference being very small, a T test shows us that H0 (”ADR of canceled and not canceled bookings do not differ”) is wrong and that ADR does have an influence on cancellations.

We have analyzed monthly cancellation rates. We can also try to see if there is a relation between it and ADR.

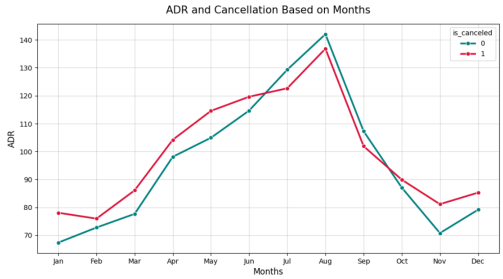


Fig. 18. Price changes do happen over the course of the year, and at summer, people seem to not cancel their pricey reservations.

Based on Fig 6 and Fig 7, canceled hotels do have a higher lead time. We also prove it by a T test. H0 (”Canceled and non canceled bookings have the same lead time”) is refused.

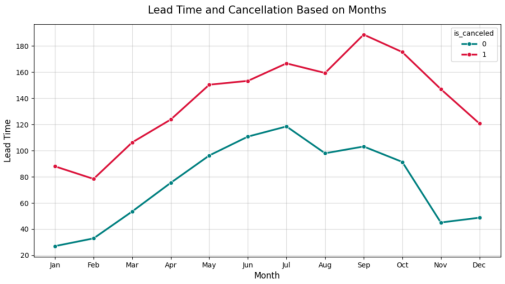


Fig. 19. Trend seems to show a higher lead time for summer. Average lead time of canceled bookings are always higher than non canceled ones.

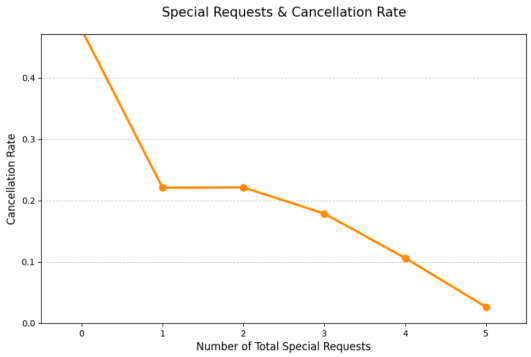


Fig. 20. There is a more psychological trend here : people who require more special requests tend not to cancel their booking.

This can be explained by how a hotel reservation becomes a commitment after having special requests. A similar pattern is also seen at required parking spaces; none of the people who required a parking space have canceled their booking. However, it should be noted that these requests are not very common in the dataset.

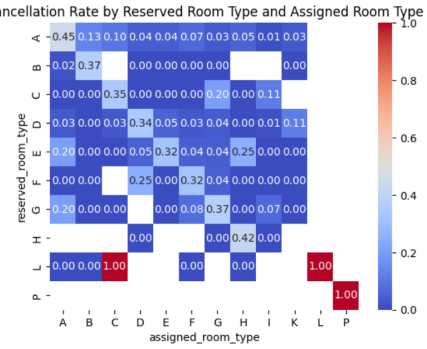


Fig. 21. At a glance, one can say : ”Cancellation rates get higher if reserved rooms are aligned with assigned rooms.” however this might also prove that most of the assigned rooms are the same as the reserved ones.

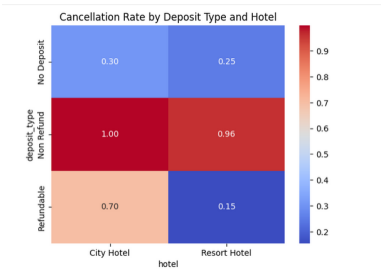


Fig. 22. There is a very strong relation between deposit type (Non Refund) and cancellation rate. This relation is independent of hotel type.

Deposit type having this much effect on cancellation rate is unexpected : especially with non-refund having a near 100 percent cancellation rate. Therefore we want to be sure if the number of non-refundables are not an outlier.

```
--- Reservation Amount ---
deposit_type    No Deposit  Non Refund  Refundable     All
market_segment
Aviation               237           0           0     237
Complementary          734           0           0     734
Corporate             4772         334           4    5110
Direct               12416          19           6   12441
Groups               10504        9172         130   19806
Offline TA/TO        19158        4991           4   24153
Online TA            56328          56          18   56402
Undefined                2           0           0       2
All                 104151       14572         162  118885
```

Fig. 23. As expected, non refund and refundable deposit types have smaller amounts of reservations. Groups is an interesting market segment, as no deposit and non refund "groups" have nearly the same amount of reservations.

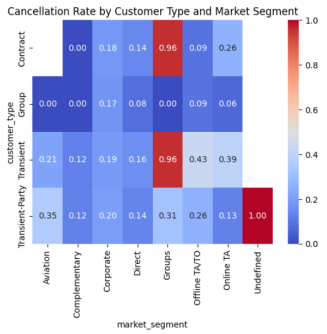


Fig. 24. Here we once again see "Groups" market segment having unusal amounts of cancellation rate. It should also be noted that undefined consists totally of canceled bookings.

IV. Conclusion

A. Summary and Future Works

This research confirms that specific attributes, such as lead time, ADR, seasonality, deposit type, market segment, special requests, and family status (an attribute that we created), play a significant role in hotel reservation cancellation rates. These factors address the primary research question of this paper, as they appear to be the most influential drivers of cancellations. Throughout this study, we first dissected each attribute individually, using visualizations to give a sense to them. We then found correlations, tested hypotheses and proved connections and differences between these variables. It is now clear that cancellations are not random occurrences but structured events.

Based on our findings, guests do not choose hotels simply because they are inexpensive. If price were the sole driver, our ADR-month graphic would be completely different : even in the most populous times people came to hotels at expensive rates. This suggests that hotels are not merely selling a bed and food; they are selling an experience. We observed that the presence of special requests significantly lowered cancellation rates. Similarly, we found that families contribute higher revenue (ADR) yet cancel less frequently. This indicates that the commitment to experiencing a holiday with loved ones keeps them from cancelling their reservations.

This paper was a great chance for the author to see how a complex system, such as hotel bookings work in intertwined data connections and how much analysis there was to uncover. While this study has uncovered significant patterns, much remains to be explored. The insights discovered here could serve as a foundation for a Machine Learning model designed to predict customer churn. Furthermore, while this paper focused specifically on cancellation rates, there are other attributes that tell us different stories, unrelated to churn analysis.

References

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